



01.07.2026

Tag 8

Von KI erstellt, von Menschen kontrolliert.

Guten Morgen!

TCP/IP

Vint Cerf 1964

IP 32Bit

$2^{32} \sim 4 \cdot 10^9$
Adressen

Internet
IANA

private IP
RFC 1918
 $\left\{ \begin{array}{l} 10.0.0.0/8 \\ 172.16.0.0/12 \\ 192.168.0.0/16 \end{array} \right\}$

VNet

DNS

April 1st RFC $\downarrow$ GitHub Repo <https://github.com/www42/a5>
~~IPv6~~

AZ-104

Learning Paths:

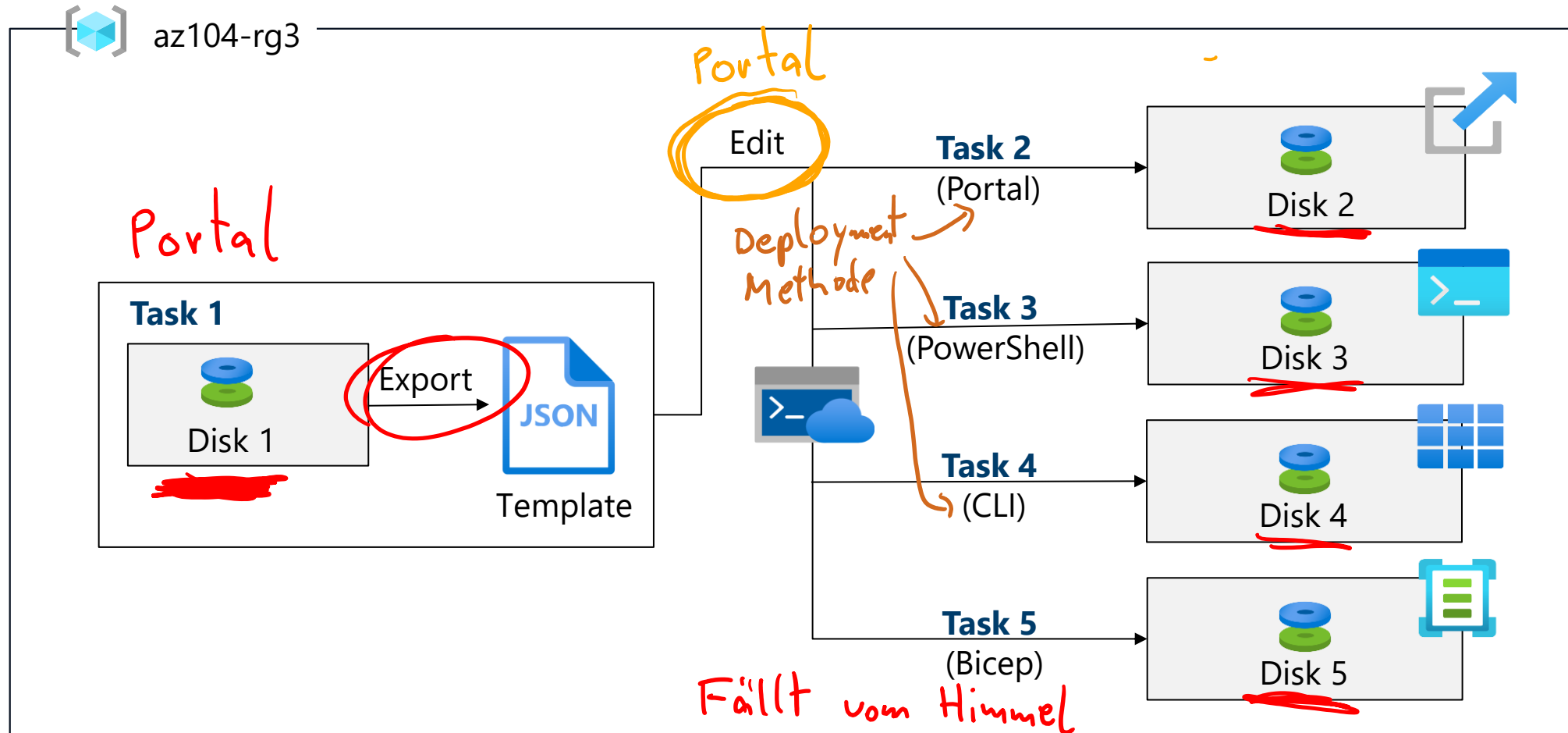
- Manage identities and governance in Azure ✓
- Prerequisites for Azure administrators – CloudShell and templates ✓
- Configure and manage virtual networks for Azure administrators
- Implement and manage storage in Azure
- Deploy and manage Azure compute resources
- Monitor and back up Azure resources

What happened yesterday



Lab 03 – Architecture diagram

VS Code
Shortcuts
Search + Replace `ctrl-H`



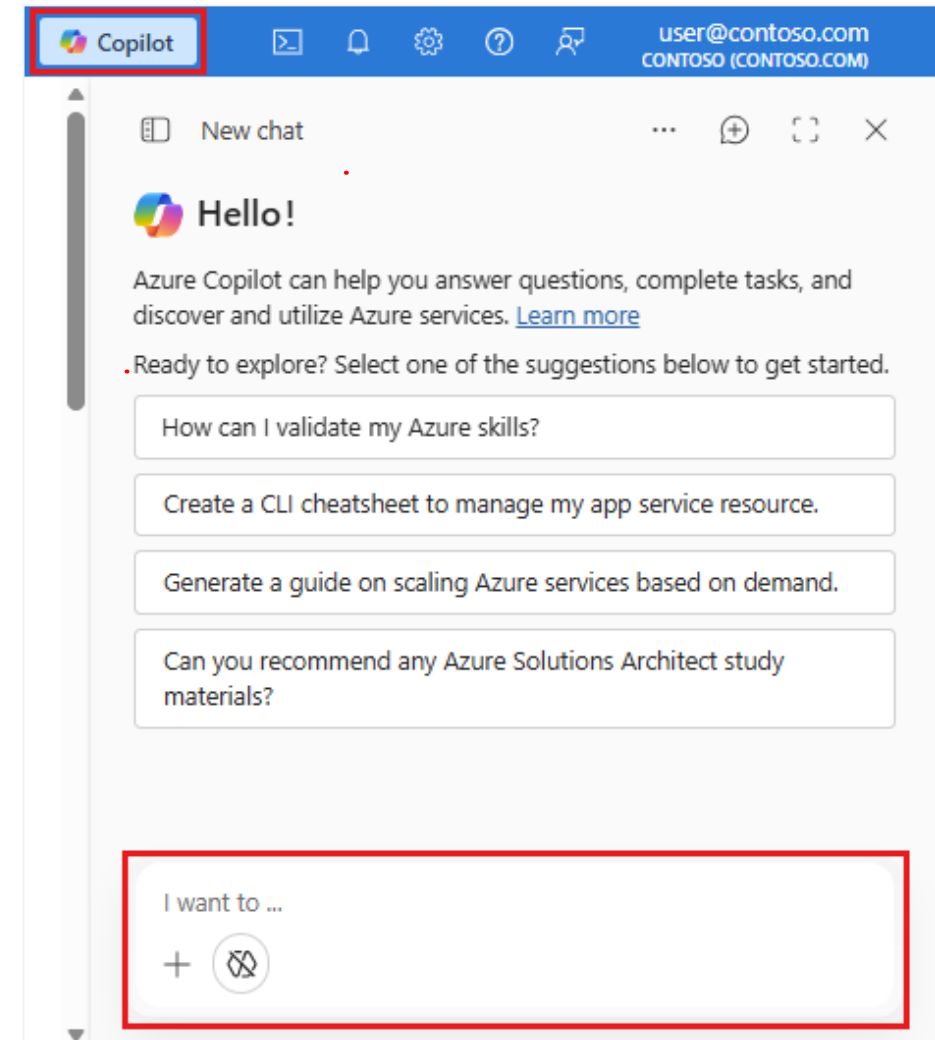
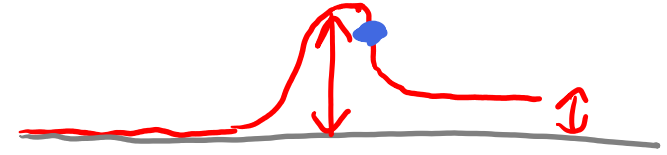
And now for something
completely different...



Microsoft Copilot for Azure

- Access in the Azure portal, through the Azure mobile app, or through AI Shell
- Helps you design, operate, optimize, and troubleshoot your Azure apps and infrastructure
- Performs many basic tasks
 - Deploy and manage virtual machines
 - Design, troubleshoot, and secure networks
 - Generate PowerShell and Azure CLI scripts
 - Analyze, estimate, and optimize costs
 - Discover and deploy workload templates

AI ?



Any sufficiently advanced technology is
indistinguishable from magic.

Arthur C. Clarke 1973



A brief history of AI

Artificial Intelligence

Machine Learning

Deep Learning

LLM
↓
Generative AI

NLP
Natural Lang Process

1950s

Artificial Intelligence

The field of computer science that seeks to create intelligent machines that can replicate or exceed human intelligence.

1959

Machine Learning

A subset of AI that enables machines to learn from existing data and improve upon that data to make decisions or predictions.

2017

Deep Learning

A machine learning technique in which layers of neural networks are used to process data and make decisions.

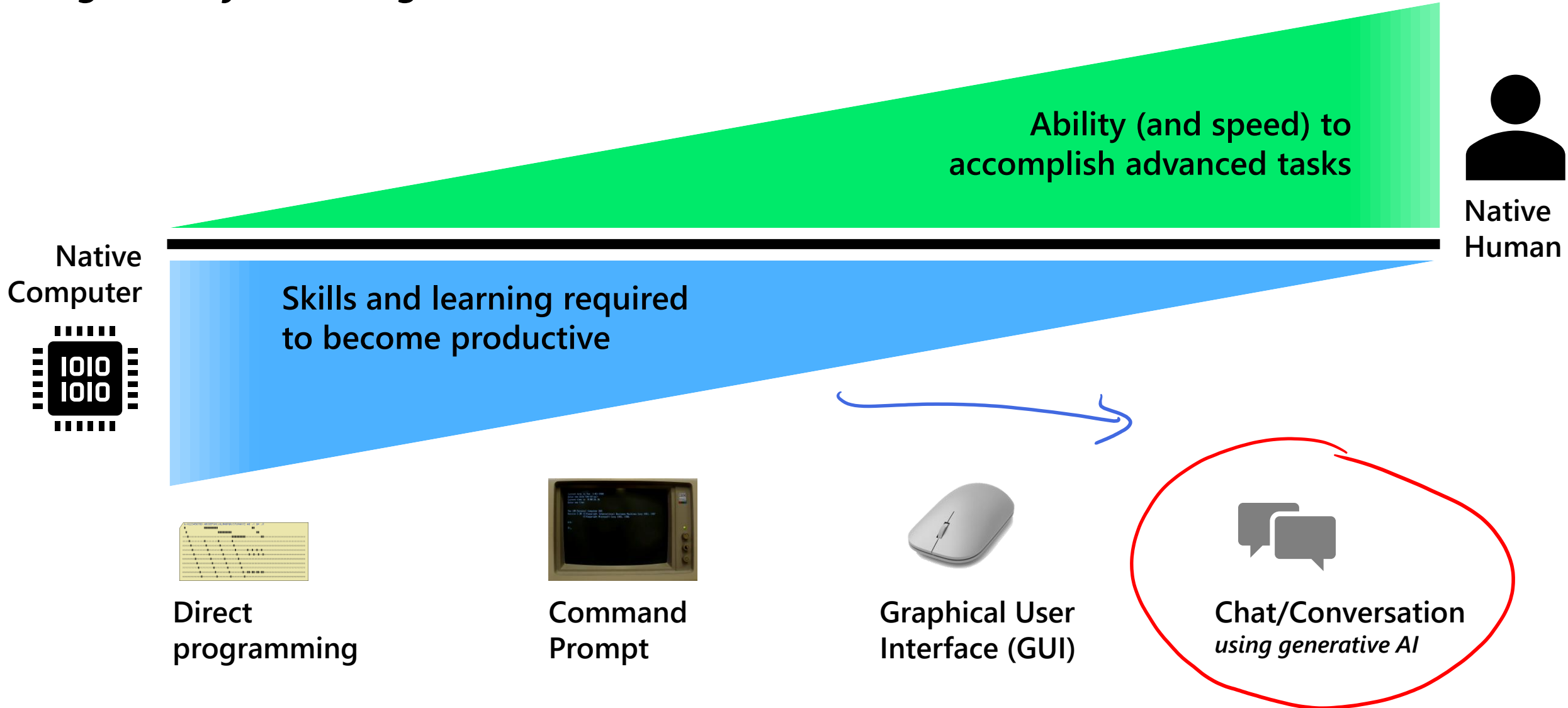
2021

Generative AI

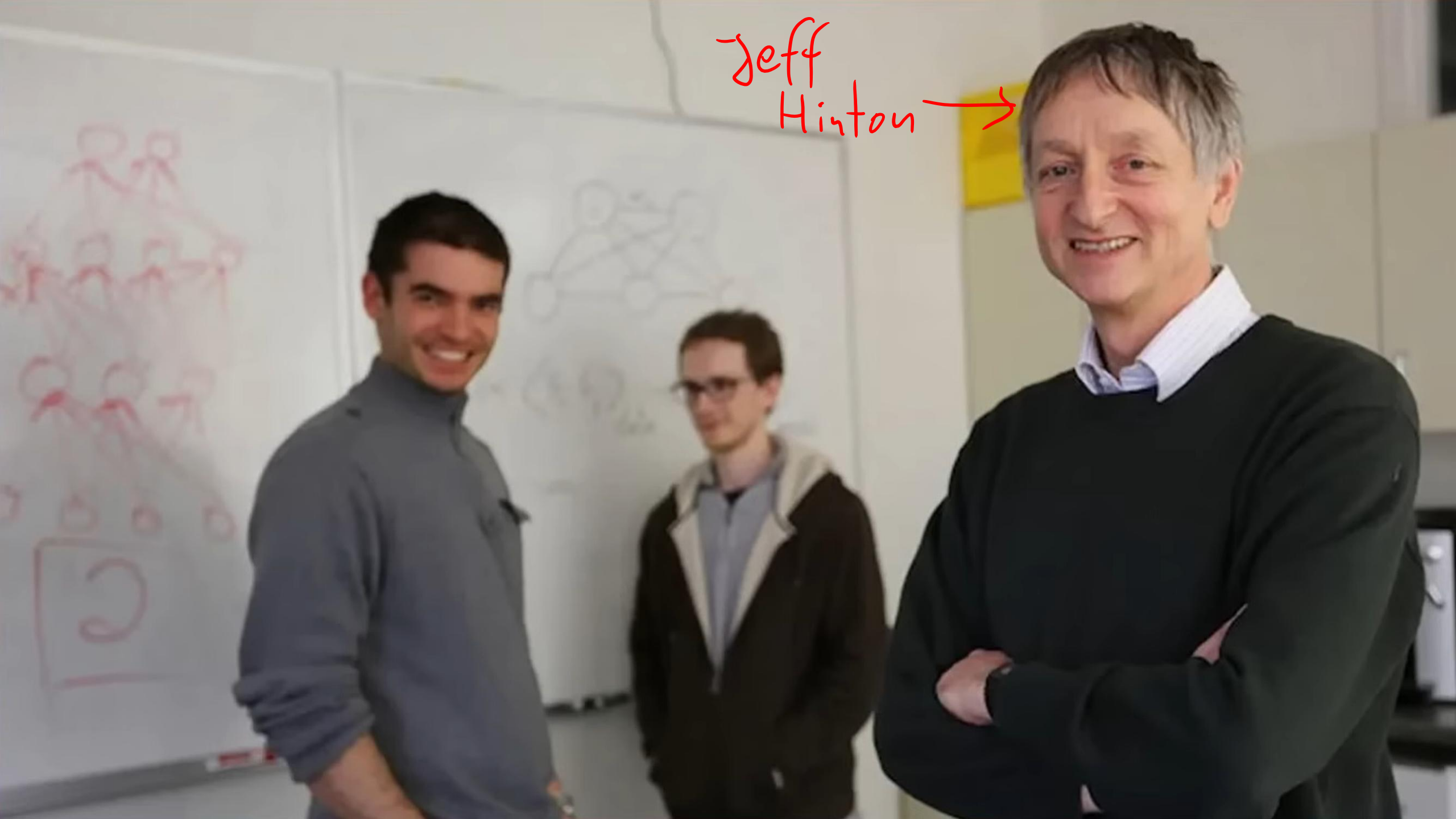
Create new written, visual, and auditory content given prompts or existing data.

Evolution of Computer Interfaces

Progressively becoming more natural/native human models



Jeff Hinton →



Introduction to generative AI and agents

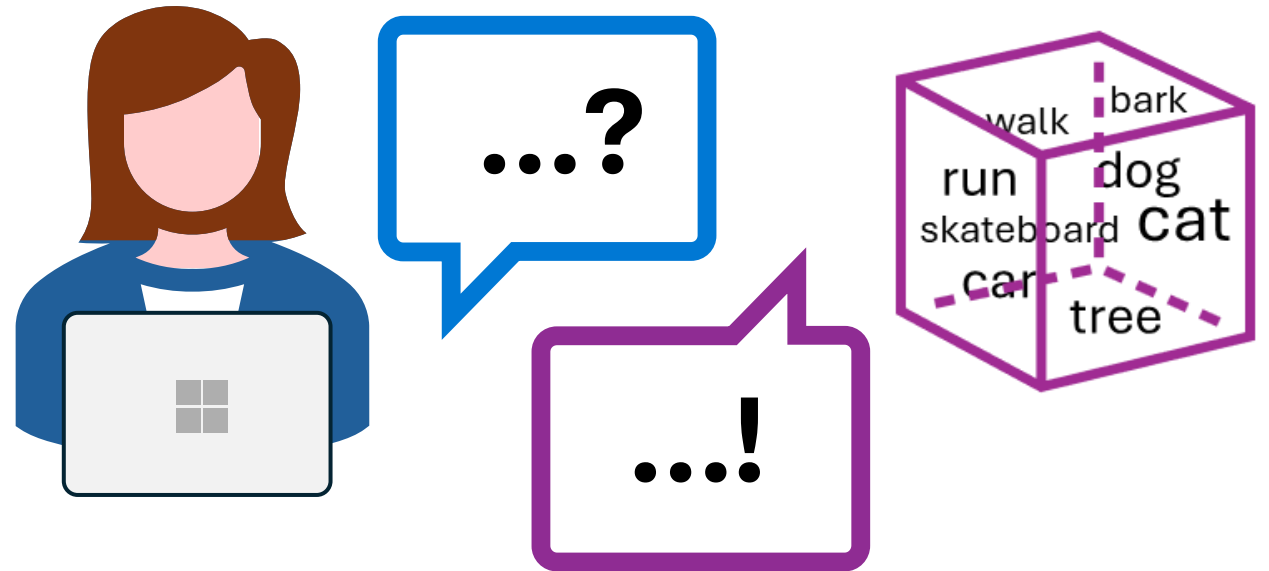
<https://aka.ms/mslearn-intro-gen-ai>



What is generative AI?

Large language models (LLMs)*

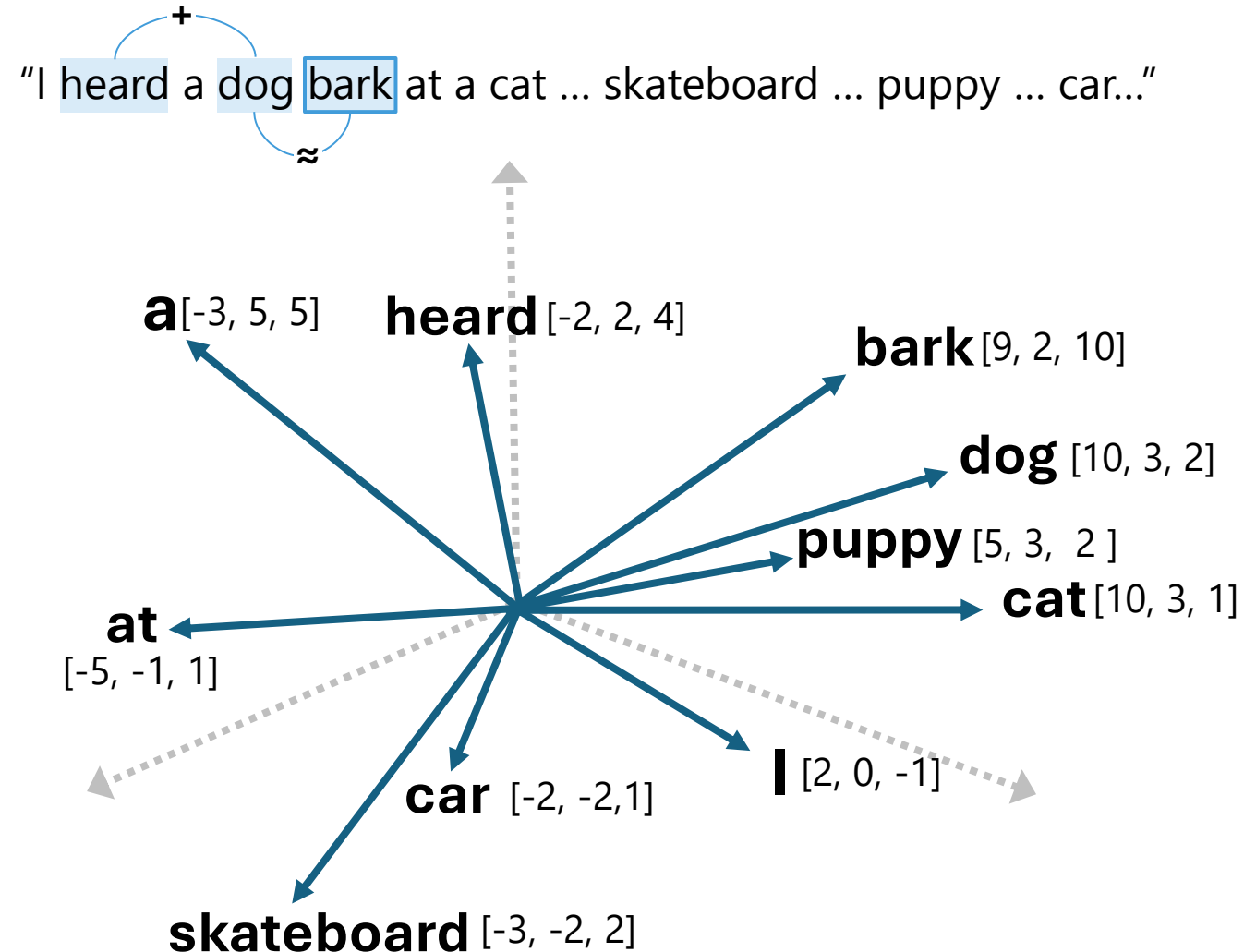
- Generate **responses** based on natural language **prompts**
- Output can be:
 - Natural language (text or speech)
 - Images (static or video)
 - Code (Python, C#, SQL, JavaScript, ...)
 - ...
- Applications of generative AI include:
 - Smart chatbots that generate responses dynamically (without hard-coded answers or logic)
 - Creative AI assistants that can produce original text, artwork, and code
 - The foundation for *agentic AI* – automation through smart, connected agents.



Generative AI models

LLMs are based on *transformer* model architecture

- Models encapsulate contextual relationships between language **tokens**
- Tokens are encoded as vector-based **embeddings**
- Embedding values are encoded using **attention** – for each token, values are based on the weighted influence of tokens around it in the training data
- Embeddings form a multidimensional vector-space
 - Semantically similar tokens have similar directions
 - Attention-based weights enable vector calculations to predict “next word” completions



What are agents?

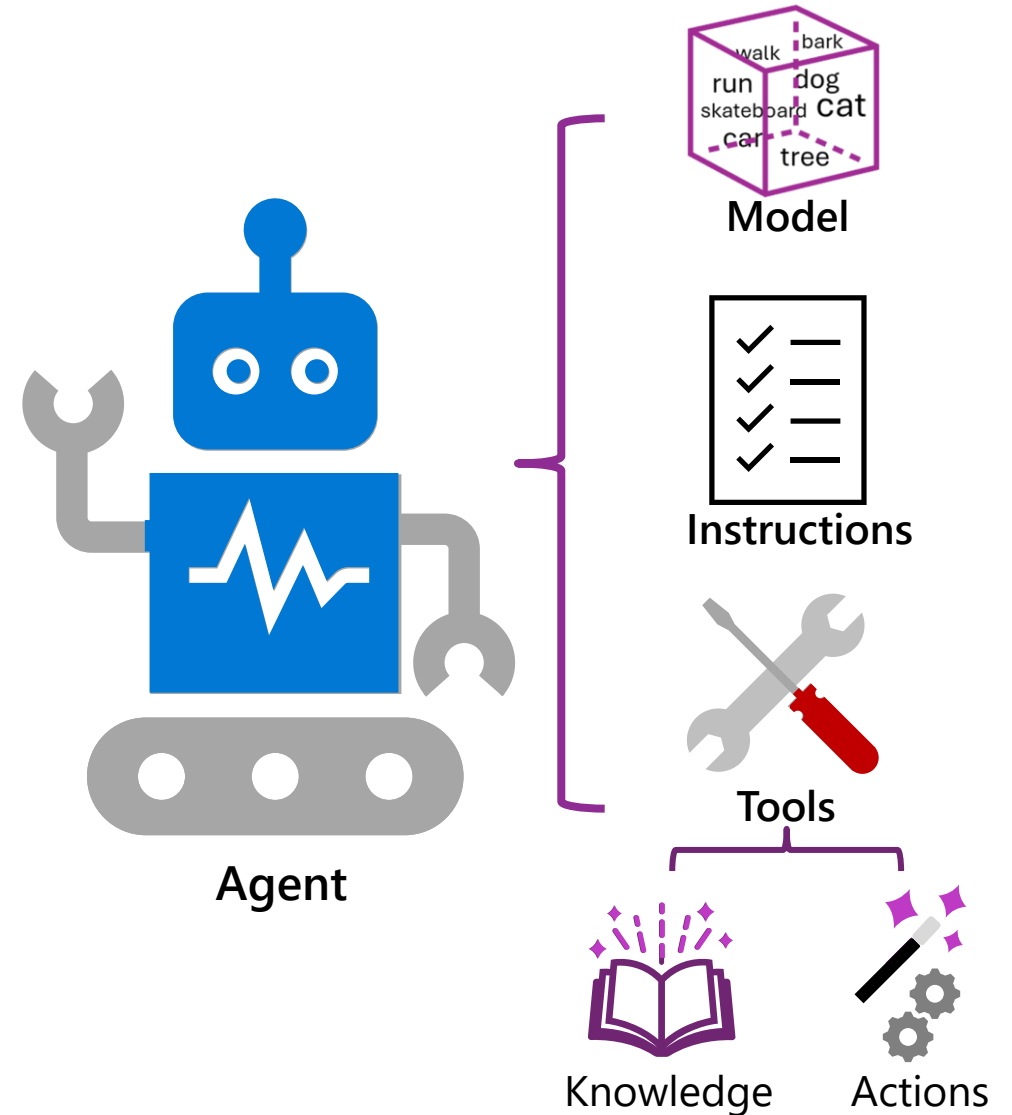
Autonomous, action-capable AI assistants

Agents use:

- A **model** for language understanding and reasoning
- **Instructions** to define behavior and scope
- **Tools** they can use to:
 - Find **knowledge** to contextualize their work
 - Perform **actions** to automate tasks

Agents collaborate with:

- Human users, to assist with tasks and processes
- Other agents, to create workflows of specialized agents, each with their own responsibilities



AZ-104

 Configure and manage virtual networks for Azure administrators



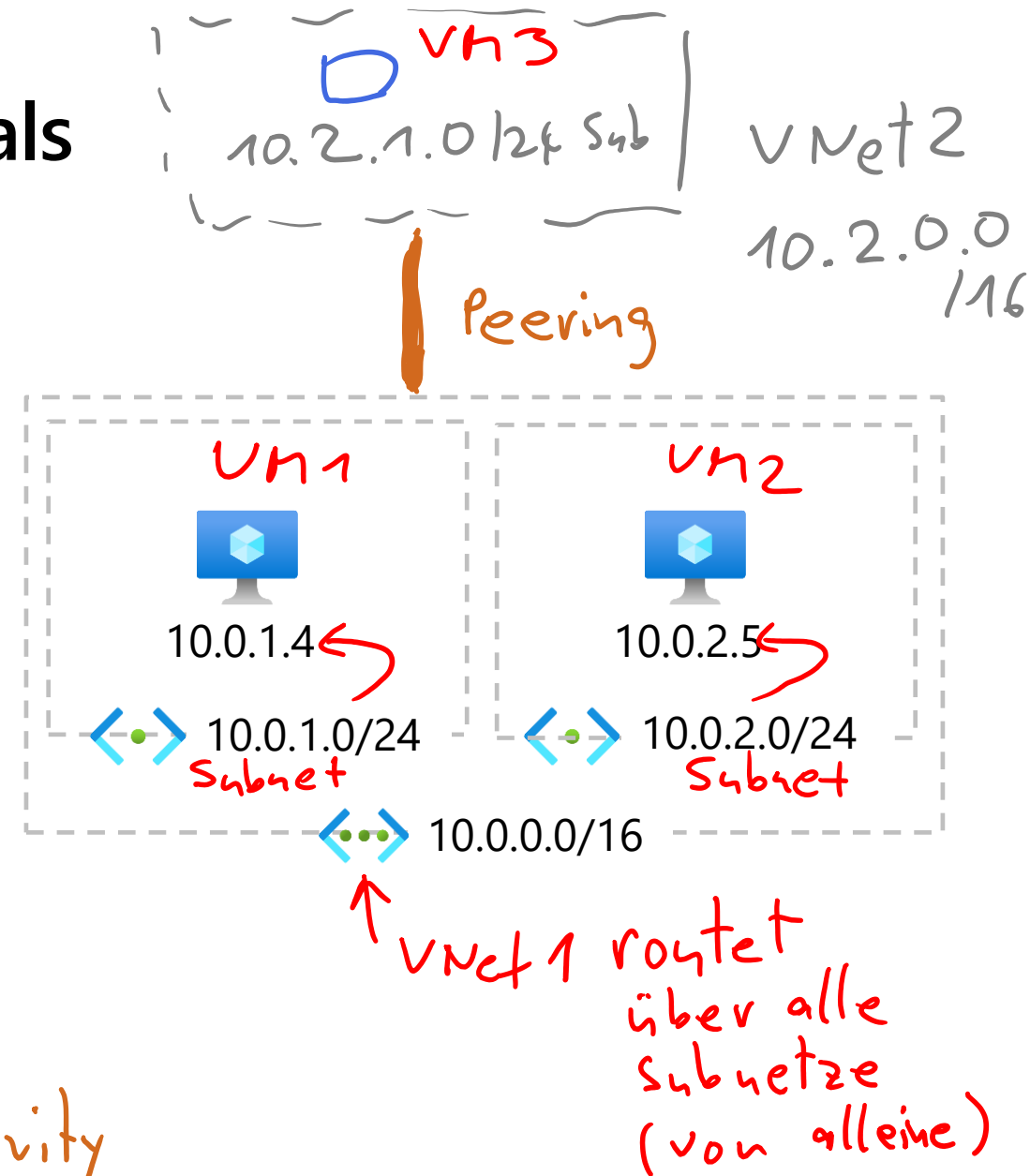
Configure Virtual Networks

VNet ~~Ulan~~

NSG Net Security Group = Firewall

Review Virtual Network Fundamentals

- **Virtual Network (VNet):** The main boundary, a large private IP address space (e.g., 10.0.0.0/16) representing your isolated cloud network.
- **Subnets:** Smaller divisions within the VNet (e.g., 10.0.1.0/24, 10.0.2.0/24) for grouping resources, often public (with gateways) or private (for databases).
- **IP Addresses:** Individual addresses assigned to resources (VMs, databases) within subnets, following the subnet's range (e.g., 10.0.1.4, 10.0.2.5).
- **CIDR Notation:** /16, /24 notation showing network size and available IP addresses.



Peering = Connectivity
Peering $\neq$ Security $\rightarrow$ N

Review IP Addressing Fundamentals

- Not all IP addresses are available for assignment.
- The first four and the last IP address in every subnet is reserved for internal platform use (network/broadcast, default gateway, Azure DNS).
- Multicast, Broadcast, Loopback, Link-Local, and Internal DNS IP address ranges are not available for assignment.
- RFC 1918 defines specific IPv4 address ranges for private networks. These addresses are not routed on the public internet.

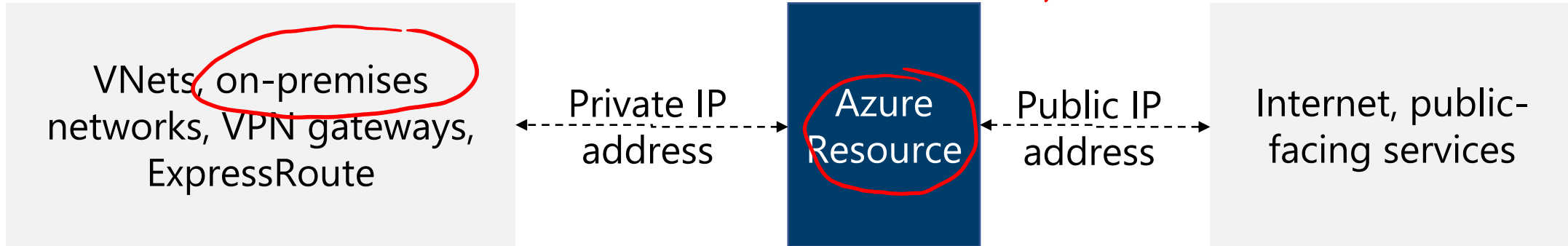
10.0.0.0/16

10.0.0.0 - 10.0.255.255 65,536 addresses

Subnets	IP address range	Size
default	10.0.0.0 - 10.0.0.255	/24 (256 addresses)

Plan for Azure IP Addressing

Hub-Spoke
Topology

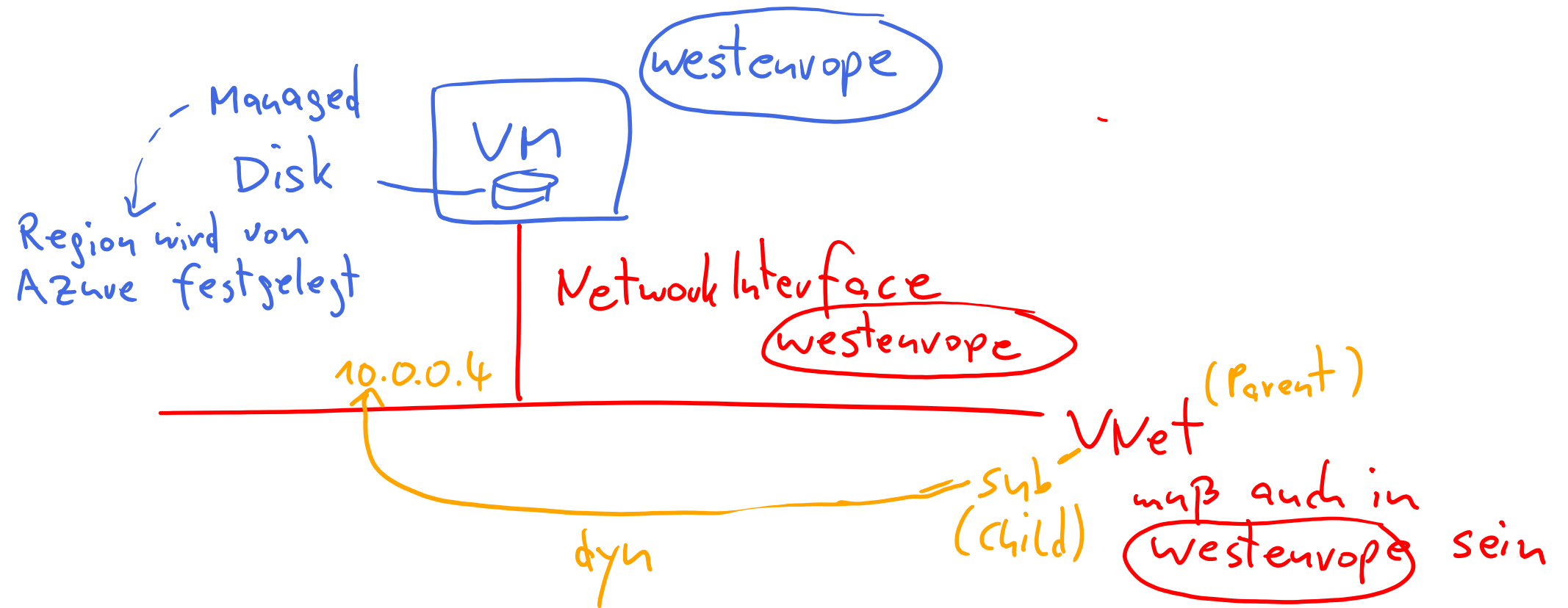


- Ensure address spaces don't overlap
- Use a few large virtual networks instead of multiple smaller ones
- Decide if your IP addresses should be private or public
- Decide if the IP addresses should be statically or dynamically allocated – consider using separate subnets
- Subnets shouldn't cover the entire address space of the virtual network
- Never manage IP addresses from within the virtual machine

Network Watcher
• Sniffer
• Topol

Warum mehrere Regions in Azure?

→ Compute und Networking in derselben Region.



Review Examples of Public IP Addressing

- Public IP addresses allow Internet resources to communicate inbound to Azure resources
- Enables communication to the internet and public-facing Azure services
- Dedicated exclusively to the resource until you unassign it or delete the resource
- Public IP address prefixes (a reserved range of public IP addresses in Azure) are also available.

Public IP addresses	IP address association
Virtual Machine	NIC
Load Balancer	Front-end configuration
VPN Gateway	Gateway IP configuration
Application Gateway	Front-end configuration
Azure Firewall	Front-end configuration

:80 HTTP
:443 HTTPS
:22 SSH
:53 DNS (Request UDP/53)
:3389 RDP

Configure Network Security Groups (NSGs)

~~DHCP~~
VPN < :443
IPSec :500

Implement Network Security Groups

nsg0
Network security group | Directory: Microsoft

Search (Ctrl+ /) <<

→ Move Delete Refresh

Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems

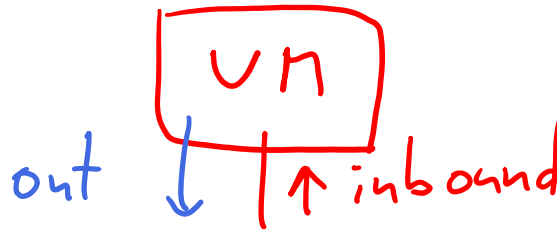
Resource group ([change](#)) : rg01
Location : East US
Subscription ([change](#)) :
Subscription ID :
Tags ([change](#)) : [Click here to add tags](#)

Custom security rules : 1 inbound, 0 outbound
Associated with : 1 subnets, 0 network interfaces

- Limits network traffic to resources in a virtual network
- Lists the security rules that allow or deny inbound or outbound network traffic
- Associated to a subnet or a network interface
- Can be associated multiple times

Determine NSG Rules

stateful



Inbound security rules

Priority	Name	Port	Protocol	Source	Destination	Action
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork	✓ Allow
65001	AllowAzureLoadBalancerInBound	Any	Any	AzureLoadBalancer	Any	✓ Allow
65500	DenyAllInBound	Any	Any	Any	Any	✗ Deny

Outbound security rules

Priority	Name	Port	Protocol	Source	Destination	Action
65000	AllowVnetOutBound	Any	Any	VirtualNetwork	VirtualNetwork	✓ Allow
65001	AllowInternetOutBound	Any	Any	Any	Internet	✓ Allow
65500	DenyAllOutBound	Any	Any	Any	Any	✗ Deny

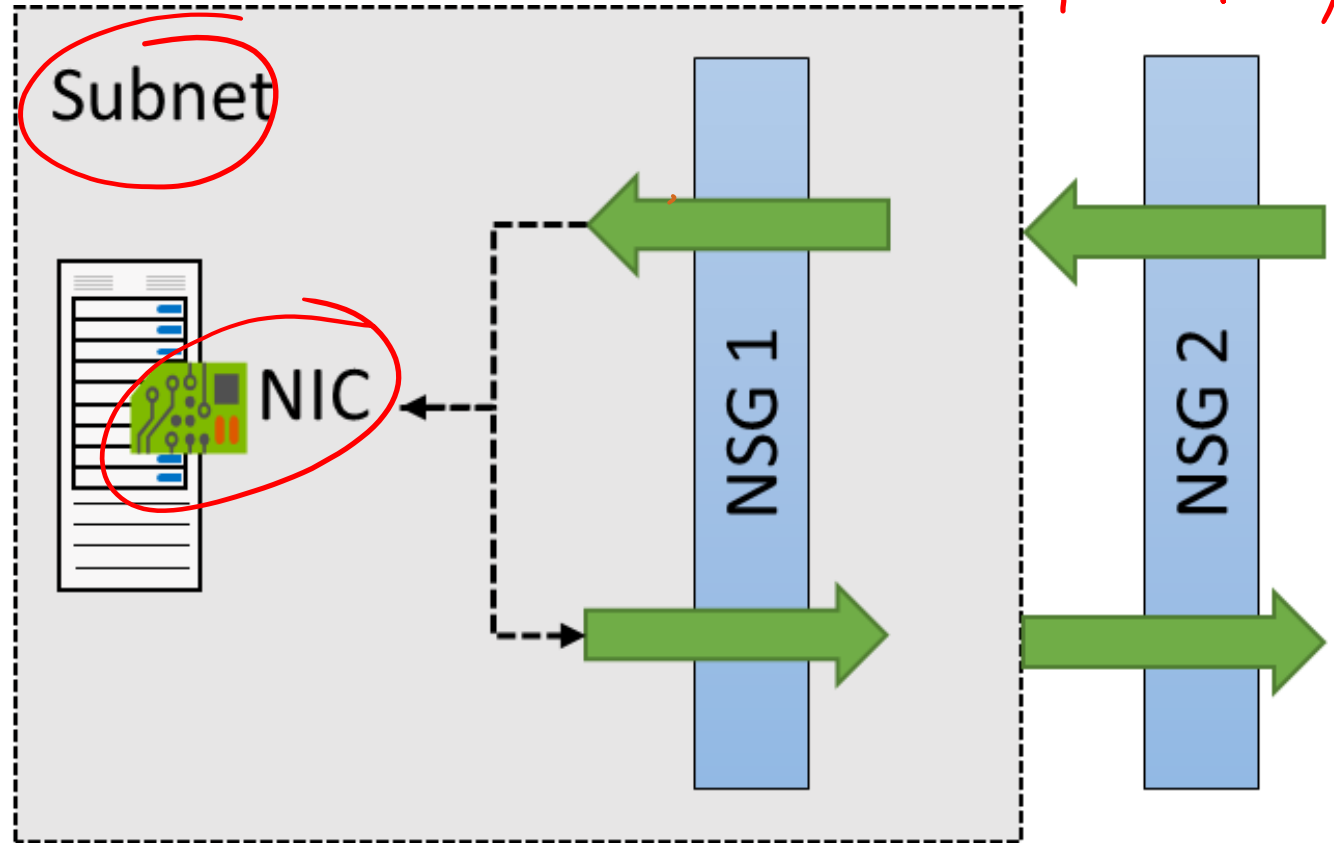
- Security rules in NSGs enable you to filter network traffic that can flow in and out of virtual network subnets and network interfaces

- There are default security rules. You cannot delete the default rules, but you can add other rules with a higher priority

Determine NSG Effective Rules

= Firewall (stateful, nur Header, nicht Body)
→ Azure Firewall \$\$, 3rd party Fw)

- NSGs are evaluated independently for the subnet and NIC
- An "allow" rule must exist at both levels for traffic to be admitted
- Use the Effective Rules link if you are not sure which security rules are being applied



Network Interface: **vm01990**

Effective security rules

[Topology](#)

Virtual network/subnet: **vnet01/subnet0**

NIC Public IP: -

NIC Private IP: **10.1.0.4**

Accelerated networking: **Disabled**

Create NSG rules

- For each rule, you can specify source and destination details, port, and protocol.
- Select from many preconfigured services, or create your own custom rules
- Rules are processed in priority order. The lower the number, the higher the priority.
- There are limits on the number of rules per NSG for each Azure location and subscription.

Add inbound security rule

Source ⓘ
Any

Source port ranges * ⓘ
*

Destination ⓘ
Any

Service ⓘ
Custom

Destination port ranges * ⓘ
8080

Protocol
☒ Any ☐ UDP
☐ TCP ☐ ICMPv4

Action
☒ Allow ☐ Deny

Priority * ⓘ
100

Name *
AllowAnyCustom8080Inbound

Service ⓘ

Custom

HTTP

HTTPS

SSH

RDP

MS SQL

MySQL

PostgreSQL

Windows Admin Center

Custom

FTP

SMTP

DNS (TCP)

DNS (UDP)

POP3

IMAP

LDAP

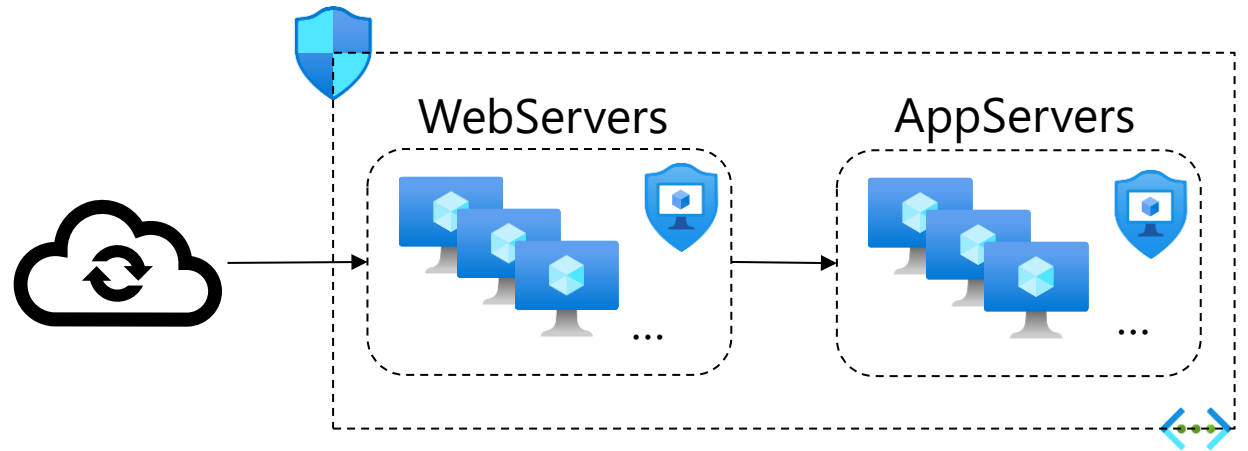
SMTPS

IMAPS

POP3S

Implement Application Security Groups (ASG)

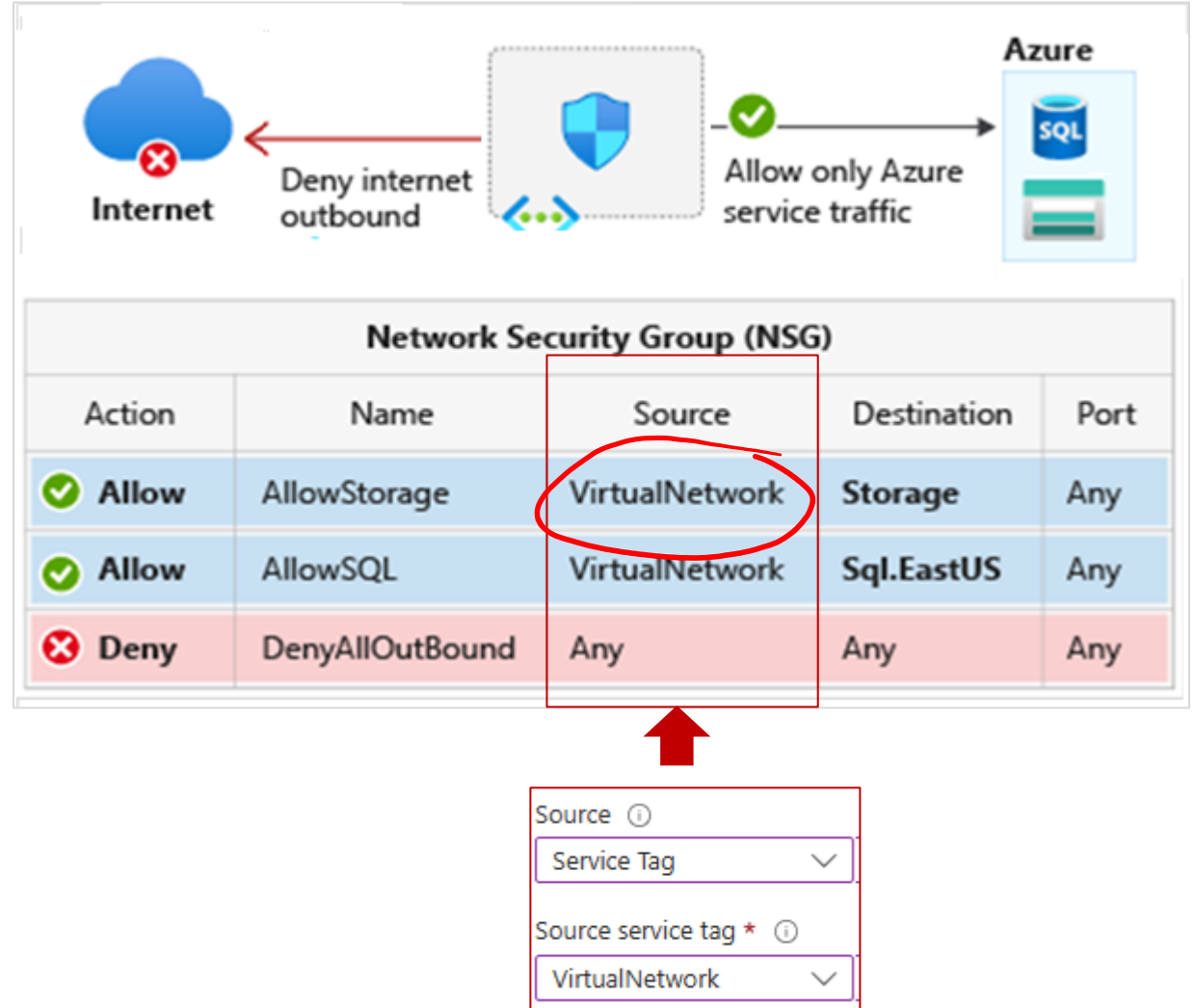
- Group VMs by application or workload and apply network security policies to those groups - instead of to individual IP addresses.
- ASGs must be in the same region and subscription as the VMs they group.
- ASGs can be the source or destination in a security rule



Action	Source	Destination	Port
Allow	Internet	WebServers	80,443
Allow	WebServers	AppServers	443
Deny	Any	Any	Any

Use Service Tags in NSGs

- Predefined groups of IP address prefixes for specific Azure services
- Many services are available including VirtualNetwork, Storage, and Internet
- Automatically updated, reduced complexity, and granular control

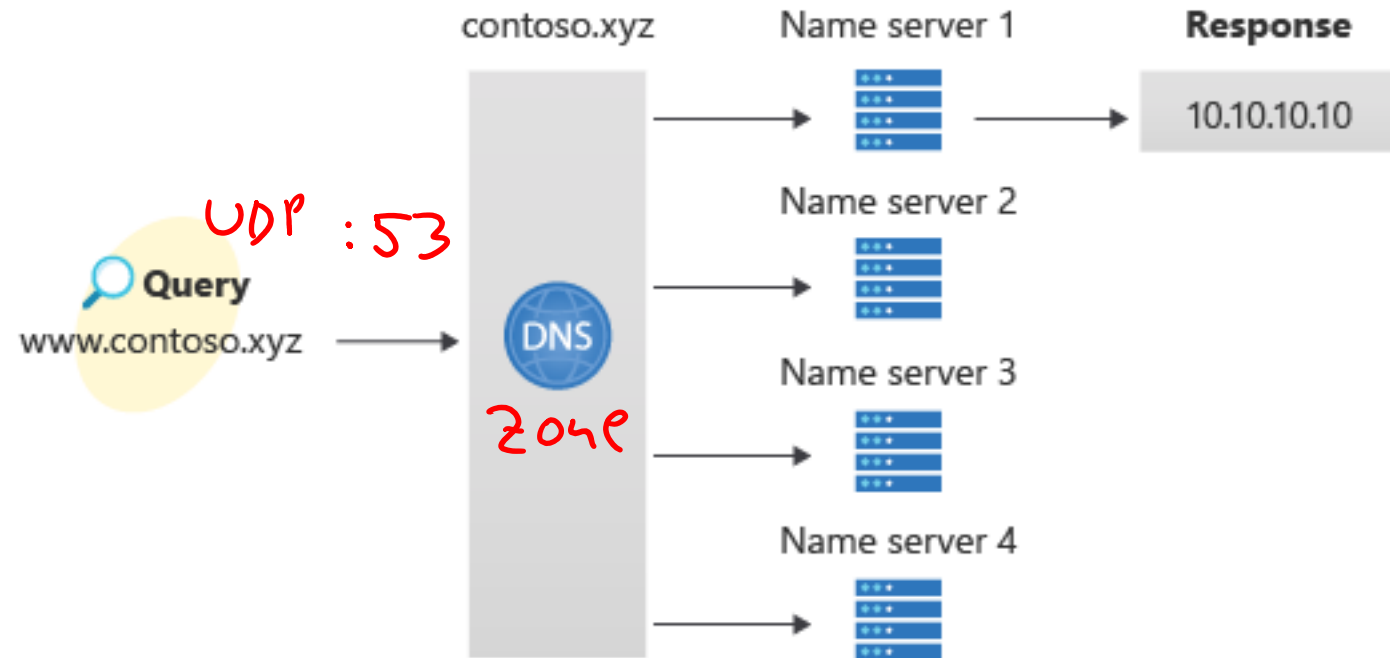


Host your domain on Azure DNS



Review Domain Name Service (DNS) Fundamentals

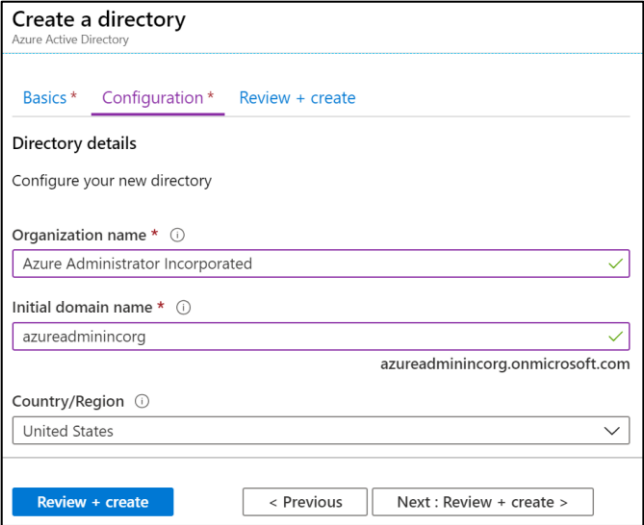
- **DNS Zone:** Hosts the DNS records for a particular domain.
- **DNS Records:** Information to handle requests.
 - A: Maps a domain name to IPv4 address
 - AAAA: Maps a domain name to IPv6 address
 - CNAME : Maps for one domain to another (alias record)
- **DNS Resolution:** The process of translating a human-readable domain name into a machine-readable IP address.



What is Azure DNS?

Tenant
♀
5erc94.onmicrosoft.com

- When you create a new tenant, a new default domain is created
- The domain has initial domain name in the form *domainname.onmicrosoft.com*
- You can add a custom domain name
- After the custom name is added it must be verified – this demonstrates ownership of the domain



Create a directory
Azure Active Directory

Basics * Configuration * Review + create

Directory details

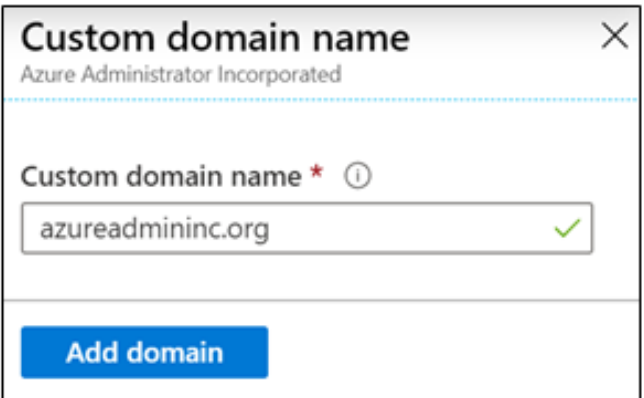
Configure your new directory

Organization name * ⓘ
Azure Administrator Incorporated ✓

Initial domain name * ⓘ
azureadminincorg ✓
azureadminincorg.onmicrosoft.com

Country/Region ⓘ
United States

Review + create < Previous Next : Review + create >



Custom domain name
Azure Administrator Incorporated

Custom domain name * ⓘ
azureadmininc.org ✓

Add domain

Create a Public DNS Zone and a DNS Record Set

You must claim and register (GoDaddy, Namecheap, or Google Domains) your domain

Create DNS zone

Basics

Tags

Review + create

A DNS zone is used to host the DNS records for a particular domain. For example, the domain 'contoso.com' may contain a number of DNS records such as 'mail.contoso.com' (for a mail server) and 'www.contoso.com' (for a web site). Azure DNS allows you to host your DNS zone and manage your DNS records, and provides name servers that will respond to DNS queries from end users with the DNS records that you create. [Learn more.](#)

Project details

Subscription *

MSDN Platforms Subscription

Resource group *

rg-dns

Create new

Instance details

Name *

azureadmininc.org

Resource group location ⓘ

East US

Review + create

Previous

Next : Tags >

[Download a template for automation](#)



Add record set

contosoces.com

Name

.contosoces.com

Type

A – IPv4 Address records

A – IPv4 Address records

AAAA – IPv6 Address records

CAA – Certificate Authorities to authorize certificates

CNAME – Link your subdomain to another record

MX – Mail eXchange records

NS – Name Server records

SRV – Service records

TXT – Text records

PTR – Pointer records

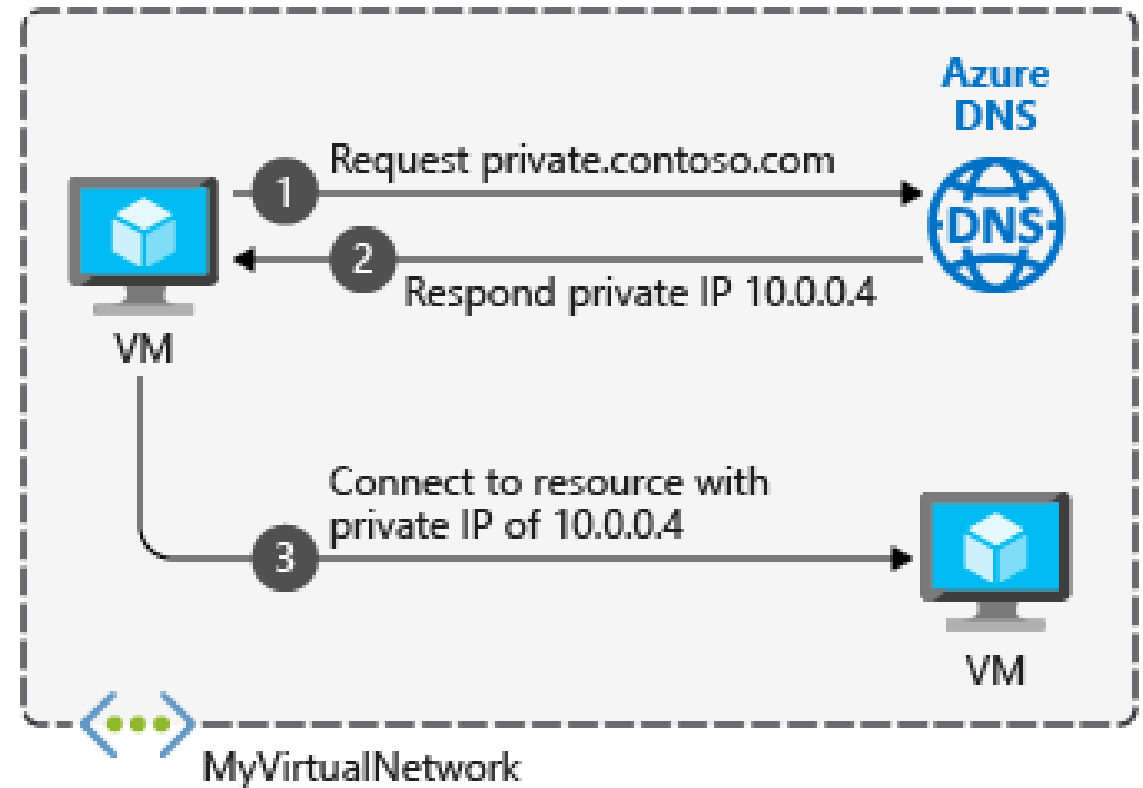
DS - Delegation Signer records

TLSA – TLS Authentication records

Configure a Private DNS Zone

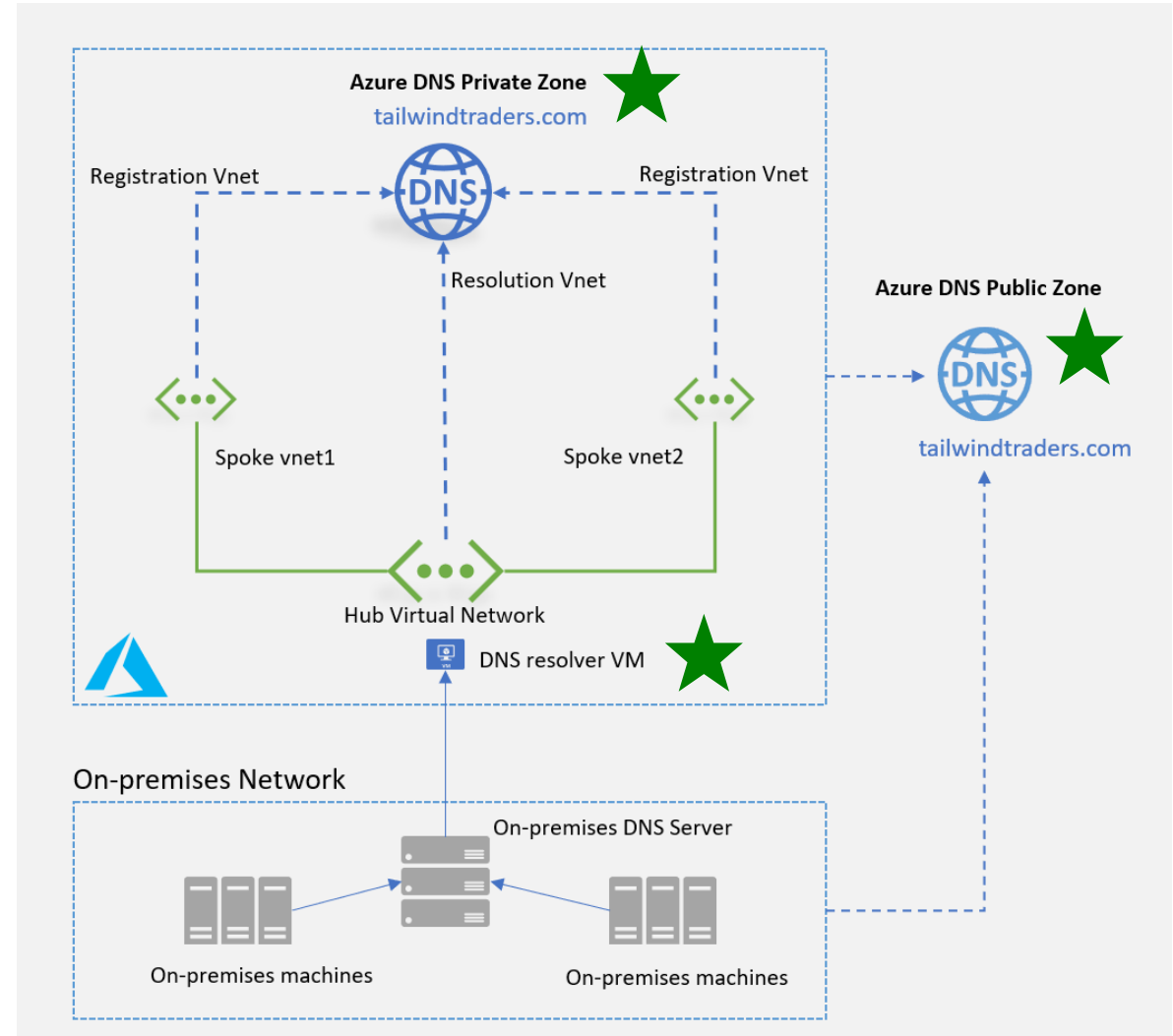
- Use your own domain names
- Access from any region, any subscription, any VNet, and any tenant
- Provides name resolution for VMs within a VNet and between VNets
- Use auto registration to populate the DNS records for VMs in the zone
- Be sure to create a link from the private zone to the virtual network

Example: Name resolution scoped to a single virtual network



Understand Azure DNS Capabilities

- **Azure Public DNS** is a hosting service that provides name resolution by using Microsoft Azure infrastructure.
- **Azure Private DNS** manages and resolves domain names in the virtual network without the need to configure a custom DNS solution.
- **Azure DNS Private Resolver** resolves DNS queries for private DNS zones from anywhere.



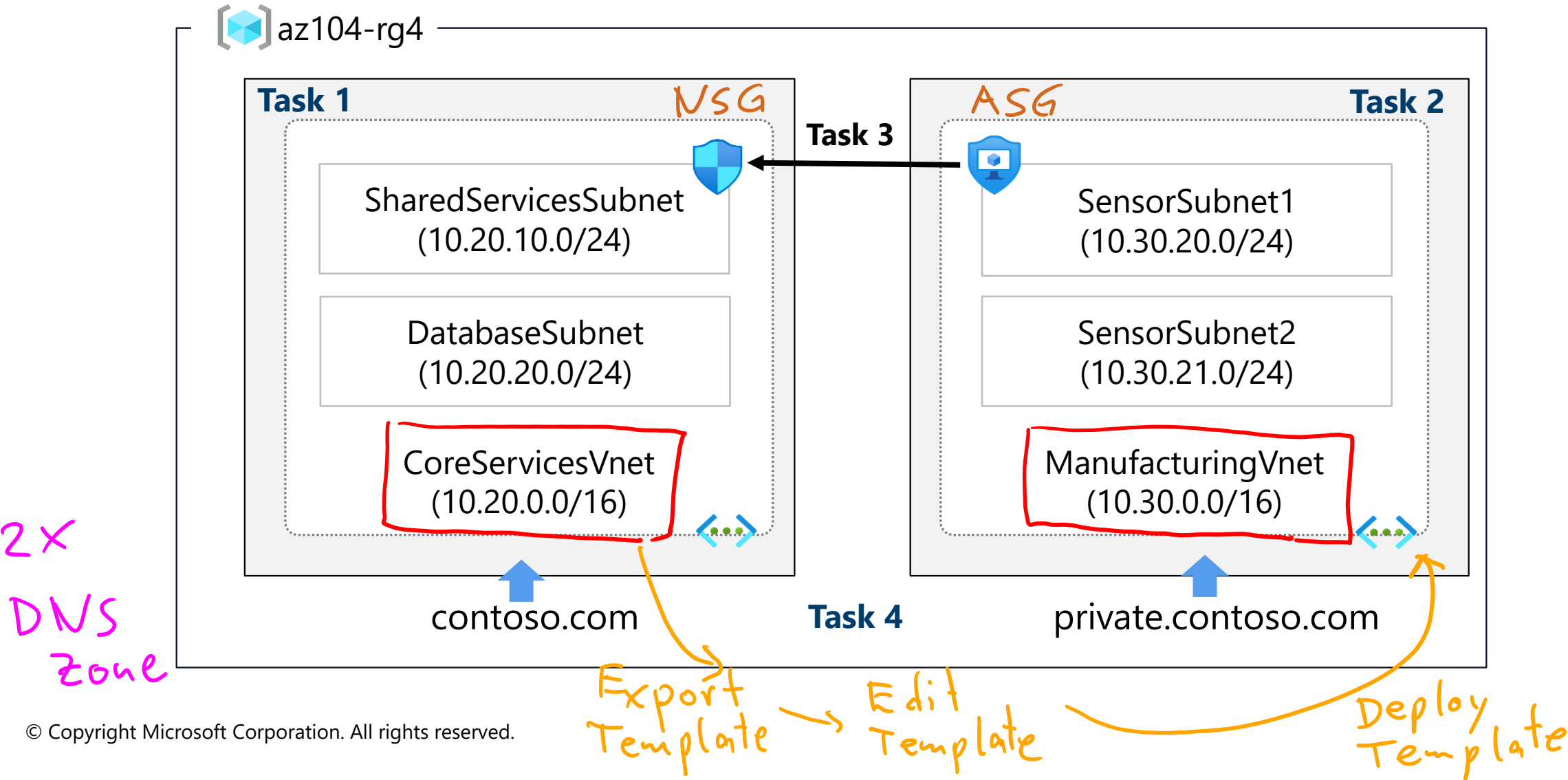
Labs



Lab 04 – Implement Virtual Networks



Lab 04 – Architecture diagram



End of presentation



